

Human: A vehicle is at the intersection if it is adjacent to a stop sign and not moving.

LEIA: I think a vehicle is beside a stop sign when it is no more than 50 inches away. Is that right?

Human: Yes.
If other vehicles arrived at the intersection before you, they have the right of way.

Before the learning started, the LEIA's lexicon included the necessary meanings of many of the individual words as well as the meanings of the constructions *I'll teach you what to do*, *a NP is a NP* (indicating the IS-A relation), *four-way intersection*, *come to a stop*, *how do I tell if*, *no more than*, and *have the right of way*.

Human: *I'll teach you what to do at a four-way stop.* The agent analyzes this sentence using the lexical sense for the construction *I'll teach you what to do* PP¹². The basic semantic interpretation of that lexical sense, which is recorded in its sem-struc, says:

- This is an instance of **TEACH** in which the speaker is the **AGENT** and the listener is the **BENEFICIARY**.
- The **THEME** of **TEACH** is the events that comprise some script, which is represented as **EVENT.SUBEVENTS**.
- That **EVENT** has some **RELATION** to the meaning of the object of the preposition—here, *a four-way stop*.

When this lexical sense is used to analyze our example, it generates the following TMR.

TEACH-1		
AGENT	HUMAN-1	; I am teaching
BENEFICIARY	LEIA-1	; you
THEME	EVENT-1.SUBEVENTS	; a new script (an EVENT with SUBEVENTS)
HAS-MP	CREATE-AND-ANCHOR-NEW-CONCEPT: OBJECT-1 CREATE-AND-ANCHOR-NEW-CONCEPT: EVENT-1	
EVENT-1		; the new EVENT "what to do at a four-way stop"
RELATION	OBJECT-1	; is related to the new object "a four-way stop"

Some notes about this TMR:

- The meaning of *a four-way stop* is underspecified—that is, it is mapped to **OBJECT**—because this is just the initial stage of learning the new expression.

- The meaning procedures (fillers of HAS-MP) serve as input to the ontology-learning function `CREATE-AND-ANCHOR-NEW-CONCEPT`; they indicate which language strings need to be converted into concepts and then incorporated into the ontology. Here, “four-way stop” will be a newly learned `OBJECT` and “what to do at a four-way stop” will be a newly learned `EVENT`.

The result of running these meaning procedures is the agent learning the following knowledge:

lexicon: the nominal sense `stop-n1`, which requires the modifier *four-way* and is mapped to the concept `FOUR-WAY-STOP`
 ontology: the concept `FOUR-WAY-STOP`, which becomes a child of `OBJECT`; and the concept `EVENT-AT-A-FOUR-WAY-STOP`, which becomes a child of `EVENT`

At this point in the process, the question is: What constitutes the first input in this dialog? Recall that the agent decides what to do with candidate knowledge only after each input, which is not necessarily a single sentence. For this scenario, we assume that the agent considers each sentence a new input, but its default decision about what to do next is wait and see. It overrides that decision if it has a burning question (illustrated below) or when learning about the given topic concludes—which may be indicated, for example, by the human asking if everything is clear.

Human: A four-way stop is a four-way intersection where all vehicles must come to a stop. The first part of this sentence indicates that the ontological parent of `FOUR-WAY-STOP` is `FOUR-WAY-INTERSECTION`. Before this learning scenario, the agent already knew the multiword expression *four-way intersection*, mapped to `FOUR-WAY-INTERSECTION`, and it understands the subsumption information thanks to a lexical sense that maps the construction *a Noun is a Noun* to the property `IS-A`. This interpretation of the polysemous word *is* is preferred thanks to a heuristic in “Disambiguate using learning preferences” that prefers subsumption analyses when the agent is in learning mode. So, the TMR for the first part of the sentence is:

```
FOUR-WAY-STOP-1
IS-A      FOUR-WAY-INTERSECTION-1
```

This information triggers a function in “Create and anchor new concept” that changes the parent of `FOUR-WAY-STOP` from `OBJECT` to `FOUR-WAY-INTERSECTION`.

The second part of this sentence, *where all vehicles must come to a stop*, expresses certain property values of a FOUR-WAY-STOP:

FOUR-WAY-STOP-1		; a four-way stop
IS-A	FOUR-WAY-INTERSECTION-1	; is a four-way intersection
LOCATION-OF	MODALITY-1	; where
MODALITY-1		
TYPE	OBLIGATIVE	
VALUE	1	; it is obligatory
SCOPE	STOP-DRIVING-EVENT-1	; to stop driving
STOP-DRIVING-EVENT-1		
THEME	SET-1	
SET-1		; all vehicles
MEMBER-TYPE	VEHICLE	
QUANT	1	

When the agent evaluates this TMR for learnable information, specific ontology-learning functions included in “Learn property values” facilitate appropriately attaching the property values in the TMR to the concept description of FOUR-WAY-STOP.

Human: You can recognize a four-way stop because it has a stop sign on each of the four corners. The lexicon includes the construction *You can recognize DirectObj because Clause*.¹³ Informally, the semantic interpretation is as follows (remember that ^ indicates “the meaning of”).

you can	recognize	DirectObj	because	Clause
MODALITY(POTENTIAL 1)	DETECT	^DirectObj	HAS-SALIENT-FEATURE	^Clause

Before considering how this applies to our reasoning-heavy example, let us consider how this construction plays out for a simpler example: *You can recognize a pug because it has a short snout*:

MODALITY-1	
TYPE	POTENTIAL
VALUE	1
SCOPE	DETECT-1
DETECT-1	
THEME	PUG-DOG-1
PUG-DOG-1	
HAS-SALIENT-FEATURE	SNOUT-1
SNOUT-1	
LENGTH	.2

This says that it is possible to recognize a pug because of the salient feature of a short snout. Our example—*it has a stop sign on each of its four*

corners—works the same way but is more complicated because it requires abductive reasoning involving the meaning of *each*, the knowledge for which is stored in the lexical sense for the construction:

Subj has [DirectObj NP [pp Prep [NP each of (Det) Num Noun]]

Interpreting the meaning of this construction requires abductive reasoning—specifically, an understanding that “a stop sign on each of its four corners” means that there are four stop signs, one on each corner.

To recap the main points about the processing of this example:

- Particular linguistic constructions suggest that particular kinds of set-oriented reasoning are needed.
- The meaning procedures encoded in the lexical senses for those constructions trigger the code that carries out that reasoning.
- The reasoning is undertaken as a part of normal TMR creation, during the fourth of the agent’s six stages of language understanding: Lexically Triggered Procedural Semantics.
- This reasoning is essential to computing the meaning of the input.
- The result of this reasoning is available in the TMR that seeds learning.
- When the agent reasons about learning ontology, a function in the “Learn property values” script incorporates this knowledge into the ontological concept FOUR-WAY-STOP.

Human: A stop sign is an octagonal red sign with white letters. Interpreting this sentence uses the lexical sense for the construction *a Noun is a Noun*, which maps to the IS-A relation and indicates a concept’s ontological parent. The modifiers *red* and *with white letters* are analyzed in the normal way, resulting in the TMR:

STOP-SIGN-1		
IS-A	SIGN-1	
SIGN-1		
SHAPE	octagonal	
COLOR	red	
HAS-OBJECT-AS-PART	LETTER-OF-ALPHABET-1	
LETTER-OF-ALPHABET-1		
CARDINALITY	>1	; a shorthand for full set notation
COLOR	white	

What is noteworthy about this TMR is that it does not directly attach the property values to STOP-SIGN; it attaches them to SIGN. As part of ontology

learning—namely, as part of the reasoning carried out by the “Learn property values” script—the agent reinterprets this TMR as the following ontological frame:

STOP-SIGN	
IS-A	SIGN
SHAPE	OCTAGONAL
COLOR	red
HAS-OBJECT-AS-PART	LETTER-OF-ALPHABET-1
LETTER-OF-ALPHABET-1	
CARDINALITY	>1
COLOR	white

This reasoning uses a straightforward rule that anticipates language inputs of the form “CONCEPT-1 IS-A CONCEPT-2+PROPERTY-VALUES” and transforms it into “CONCEPT-1+PROPERTY-VALUES IS-A CONCEPT-2.” It is the latter knowledge that the agent adds to its ontology.

Human: It is usually about 30 inches across. But it can also be 35 inches wide. The first thing to note about these sentences is that they include details that most people neither know nor would think to mention. However, to optimize an agent’s learning in a high-stakes domains like driving, human teachers would be well advised to be as specific as possible. In other domains, relevant scalar values will be common knowledge, making the type of reasoning illustrated by this example widely applicable.

The first and second sentences require different kinds of reasoning that are carried out at different points in the learning process. The first one requires interpreting the meaning of *about* when it modifies a scalar attribute.¹⁴ A meaning procedure attached to the relevant lexical sense of *about* translates the named value into a range surrounding it. The default rule, which is used for objects that are not provided with more specific rules, involves generating a range using 6 percent of the listed value. Here, the result is the range 28.2–31.8 inches. So, after the first sentence, the agent has the following knowledge to seed learning:

STOP-SIGN	
WIDTH	28.2–31.8 (MEASURED-IN INCH)

The next sentence indicates that the width can be 35 inches, resulting in the following knowledge:

STOP-SIGN	
WIDTH	35 (MEASURED-IN INCH)

The agent’s “Learn property values” script includes a function to deal with such discrepancies. It learns the WIDTH of a STOP-SIGN as a tuple listing the evidence count, minimum and maximum known values, and the mean: {2, 28.2, 35, 31.6}.

Human: After you stop at a four-way stop, you must decide when to start driving. This sentence presents an ordered pair of events that includes a necessary precondition. Language understanding presents no special challenges and results in the following TMR:

STOP-DRIVING-EVENT-1		; stopping driving
LOCATION	FOUR-WAY-STOP-1	; at a four-way stop
PRECONDITION-OF	MODALITY-1	; is a precondition of
MODALITY-1		; having to decide
TYPE	OBLIGATIVE	
VALUE	1	
SCOPE	DECIDE-1	
DECIDE-1		
AGENT	LEIA-1	
THEME	ASPECT-1.TIME	; when to start driving
ASPECT-1		
PHASE	begin	
SCOPE	DRIVE-1	

When the agent evaluates this TMR using the “Learn property values” script, it reinterprets the PRECONDITION-OF property as conveying the next filler of the SUBEVENTS property of EVENT-AT-A-FOUR-WAY-STOP script. As earlier, this is a simple mapping rule that recognizes a particular shape of a TMR and transforms it into something more suitable for ontology learning.

Human: If there are no vehicles at the intersection, you can go. This input is straightforward in terms of language understanding, resulting in the following TMR:

MODALITY-1		
TYPE	PERMISSIVE	
VALUE	1	; you can
SCOPE	DRIVE-1	; drive
PRECONDITION	LOCATION-1	; if
LOCATION-1		; at the intersection
DOMAIN	INTERSECTION-1	
RANGE	SET-1	; there are no vehicles
SET-1		
MEMBER-TYPE	VEHICLE	
CARDINALITY	0	